

- **COMP 2522**
- **Fall 2025**
- **Object Oriented Programming 1**

Quiz 3/ Version 1

1. **Why would you use a final class? What does it mean if a class is final? (2 marks)**
2. **Why would you use a final method? What does it mean if a method is final?(2 marks)**
3. **Explain everything that is wrong with the following code. Circle any errors and explain what is wrong and how to fix it. Use best practices(but ignore JavaDoc here). (8 marks)**

Quiz 3/ Version 2

1. **What is the rule when overriding hashCode() in Java?(1 mark)**
2. **Create a class Phone with a yearMade variable. Make a getYearMade() method that can not be overridden. Extend the class with an iPhone class that cannot be extended. iPhones have a String colour variable. Phones made in the same year are equal. But iPhones with the same colour are equal.(11 marks)**
3. **Explain everything that is wrong with the following code. Circle any errors and explain what is wrong and how to fix it. Use best practices(but ignore JavaDoc here). (8 marks)**

```
abstract final class A extends Object
{
    Public private final int x;
    A(final int x)
    {
        This.x = x;
    }
    final void y()
    {
        System.out.println("this is x: " + x);
    }
    int getX()
    {
        return x;
    }
}
```

```
Class B extends A
{
    B(final String z)
    Private final String z;
    {
        super(z.length());
        this.z = z;
    }
    @Override
    Void y()
    {
        System.out.println("hi");
    }
    private boolean equals(final Object o)
    {
        Return this.x ==x;
    }
}
```